



Configuring Cross-Network External Border Gateway Protocol (eBGP)

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Purpose

The ultimate purpose of this lab will be the configuration of multiple areas, that are connected to a central backbone area using eBGP. We will be looking for the success in the redistribution of routes on edge routers that are not directly connected. This lab will help with understanding communications between cell service providers or ISP's and how you can make use of different routing protocols to make interconnected networks.

Background Information

BGP

Border Gateway Protocol (BGP) was created in 1989 to address the need for more efficient and scalable methods of routing between different isolated provider networks otherwise known as **Autonomous Systems** (ASes) on the internet. Autonomous systems are a collection of IP networks and routers under the control of single entity or organization that presents a common route to the internet. Before BGP, there was a protocol named Exterior Gateway Protocol (EGP) that was used, however it was not suitable for the scale and complexity of the rapidly developing internet. As the internet made many leaps in succession during the 1980s, there was a need for a protocol that could be scalable; the internet was growing rapidly, and the number of networks (autonomous systems) was increasing, necessitating a more scalable routing solution that could calculate on an international scale. As well as include policy-based routing as organizations wanted more control over how traffic is routed between ASes and on their own internal network, not just based on the shortest path or best link, but also on business, administrative, and policy considerations. It was also required to be stable as the internet needed a protocol that could adapt to changes (such as link failures) and prevent routing loops or instabilities that could cause major outages. Finally, its greatest challenge needed interdomain routing, as the internet expanded to continental connection, routing had to be done between different ASes (interdomain), not just within a single AS (intradomain). This is the core trait of BGP a program designed for expansion. Thus, BGP was introduced in 1989 with the release of BGP-1 by Yakov Rekhter and many others, later enhanced into BGP-4 (the version in use today) with the inclusion of support for Classless Inter-Domain Routing (CIDR) and other improvements.

How BGP Works

BGP is a path vector protocol designed to mediate exchanges of routing information between different ASes. It is the backbone protocol used by all major ISP's (ISPs and large enterprises) to decide the best path for data to travel across local and global internet. Some key features of BGP include, but are not limited to: Inter-AS routing, path vector routing policy, and loop prevention.

Inter-AS Routing allows routers in one AS to exchange routing information with routers in other ASes. Each AS is identified by a unique Autonomous System Number (ASN). BGP uses a path vector algorithm, meaning it keeps track of the full path of ASes that routing information has passed through. Each route advertised by BGP includes a list of ASes that the packet will

traverse to reach the destination, also known as the AS path. Routing Policies allows network administrators to define rules and limitations based on various attributes (such as AS path, prefix length, or other metrics) unlike other protocols. As BGP uses the AS path as a loop prevention mechanism. If a router receives a route advertisement that contains its own AS number in the AS path, it will reject that route to avoid network loops.

BGP Path Selection Algorithm

When BGP routers need to select the best route to a destination, they use a set of rules or attributes to determine which path to choose, the BGP rules and attributes are as follows. To prefer the path with the **highest local preference**: Local preference is a BGP attribute used to influence routing decisions within an AS. It is manually configured by network administrators and can be used to prefer one exit connection over another. To prefer the path with the **shortest AS path**: The AS path is a list of ASes the route has passed through. A shorter AS path is preferred as it indicates fewer hops and, in theory, a more direct route. To prefer the path with the **lowest origin type**: BGP assigns an origin code to each route, indicating whether the route was learned from an Interior Gateway Protocol (IGP), an Exterior Gateway Protocol (EGP), or manually configured. Routes originating from an IGP are preferred over EGP, and EGP is preferred over manually configured routes. To prefer the **path with the lowest MED** (Multi-Exit Discriminator): The MED is an attribute used to influence routing decisions between two ASes. It's used to suggest to a neighboring AS which of several possible entry points to choose, and the path with the lowest MED value is preferred. To **prefer eBGP over iBGP**: If a route is learned from an eBGP (external BGP) peer (a peer in another AS), it is preferred over a route learned from an iBGP (internal BGP) peer (a peer within the same AS). To prefer the path with the lowest IGP metric: The next step involves checking the IGP metric (such as OSPF or IS-IS) to the next hop. The path with the **lowest IGP metric** to the next hop is preferred. Tie-breaking: If all the previous steps result in ties, BGP will choose the path that was received first (i.e., the earliest in time).

eBGP

eBGP (External Border Gateway Protocol) is a path-vector routing protocol that exchanges routing information between different autonomous systems (ASes) linked through the Internet. A program developed to address the growing need for scalable, flexible, and policy-driven inter-domain routing, eBGP enables communication between independent networks under different administrative control, such as Internet Service Providers (ISPs). eBGP routers share routing information through BGP's protocol update messages, which include important attributes such as the AS path, prefix, next-hop IP, and various efficiency/speed based metrics that influence routing decisions. The AS path attribute plays a critical role in preventing routing loops by ensuring that routes are only accepted if they do not traverse the AS that is advertising them. eBGP allows network operators to define routing policies using attributes like local preference, MED (Multi-Exit Discriminator), and AS path before controlling the flow of traffic between ASes based on business needs, such as optimizing performance, increasing redundancy, or decreasing costs. It is fundamental to the global Internet infrastructure, supporting everything from large-scale ISP backbone routing to multi-homing for enterprises, and enabling precision control over how traffic is routed across diverse networks. This is important to note that eBGP is

not for local use and is used majorly as an interconnecting bridge between internet suppliers and companies. Other much faster much more detailed protocols like OSPF, EIGRP, or IS-IS should be used on small home and company networks for increased speeds as the BGP's bandwidth is not needed.

BGP was created to address challenges of routing between different ASes and to provide a flexible, scalable, and policy-driven approach to internet routing. It uses a path vector protocol and a rich set of attributes and decision-making rules to select the best route for traffic. As one of the backbone protocols for interdomain routing, the protocol plays a crucial role in maintaining stability, scalability, and flexibility of the internet.

EIGRP

EIGRP (Enhanced Interior Gateway Routing Protocol) was developed by, and is proprietary to, Cisco Systems. In the early 1990s it was meant to overcome the limitations of older hop metric, no calculation, simplistic routing protocols like RIP. It was designed to address the need of companies and agencies to have a more scalable, efficient, and faster-converging routing protocol suitable for large, complex enterprise networks and facilities. Unlike RIP, which is limited by a 15-hop maximum distance and periodic full route-state updates, EIGRP uses a hybrid approach, combining features of both distance-vector (speed and hop count) and link-state protocols. It uses the Diffusing Update Algorithm (DUAL) to ensure loop-free and optimal route selection, while also supporting differential or partial updates, meaning that only changes in the network topology are sent, reducing network bandwidth usage better optimizing existing infrastructure. EIGRP's routing decisions are based on a composite metric that calculates bandwidth, delay, reliability, and load, allowing it to choose the best and fastest, path to a destination with more precision than RIP, which relies on hop count alone. Additionally, EIGRP supports fast convergence and simple scalability, making it ideal for large networks. It also minimizes disruptions by quickly recalculating the best paths when network topology changes occur. Typically, EIGRP is used in enterprise networks where multiple routers need to communicate dynamically and efficiently without human intervention, making it the go-to protocol for Cisco-based infrastructures.

OSPF

Open Shortest Path First (OSPF) is a widely used interior gateway protocol (IGP) designed to efficiently and dynamically manage the routing of data within an autonomous system (AS). Initially developed in the late 1980s and standardized by the Internet Engineering Task Force (IETF) in RFC 2328, OSPF has become a cornerstone of modern networking, providing a scalable, reliable, and flexible solution for routing in large and complex networks as well as small home office configurations. OSPF operates based on the link-state routing algorithm, which allows routers to build an up-to-date, complete map of the entire network topology and make routing decisions based on this map's metrics.

Development and Creation of OSPF (OSPF 1, OSPF v2, Dijkstra's Algorithm, Ipv6 OSPF v3)

The development of OSPF can be traced back to the late 1980s when the need for a more efficient, scalable, and flexible routing protocol for IP networks became evident. The first version of OSPF, **OSPFv1**, was developed under the IETF's Routing Area Working Group in response to the **limitations of the Routing Information Protocol (RIP)**, which had significant issues in larger networks, such as slow convergence times, limited scalability, and reliance on hop counts as the sole metric for determining optimal path leading to network congestion and inefficiency that could have been otherwise avoided.

OSPF was designed to address these limitations, using a more sophisticated **link-state routing algorithm**, which allowed routers to share information about the state of their links (for instance, their interfaces and connections to other routers). This was a significant improvement over RIP, which only communicated hop counts and **could not effectively represent more complex network topologies**.

The development of **OSPF v2** (Standard ipv4 OSPF), the version that is widely used today, was formally completed in 1991 and standardized in RFC 2328. This version introduced numerous improvements, such as support for large networks, the ability to scale better in both large-scale enterprise networks and service provider environments, and a more detailed approach to managing route metrics, including the introduction of the **Dijkstra algorithm** to calculate the shortest path from the recorded link states and metrics.

Dijkstra Algorithm:

Dijkstra's algorithm is an algorithm for finding the shortest paths between nodes in a weighted graph, which may represent, for example, road networks. It was conceived by computer scientist Edsger W. Dijkstra in 1956 and published three years later.

Dijkstra's algorithm finds the shortest path from a given source node to every other node.: 196–206 It can also be used to find the shortest path to a specific destination node, by terminating the algorithm once the shortest path to the destination node is known. For example, if the nodes of the graph represent cities, and the costs of edges represent the average distances between pairs of cities connected by a direct road, then Dijkstra's algorithm can be used to find the shortest route between one city and all other cities. A common application of shortest path algorithms is network routing protocols, most notably IS-IS (Intermediate System to Intermediate System) and OSPF (Open Shortest Path First). It is also employed as a subroutine in other algorithms such as Johnson's algorithm.

As networking technologies and requirements continued to evolve, the need for IPv6 support grew, leading to the creation of **OSPFv3**. This version, standardized in RFC 5340, incorporated improvements and **adjustments for IPv6**, allowing OSPF to operate in modern IP-based networks **supporting both IPv4 and IPv6** protocols.

OSPF Usage

OSPF is used in a variety of networking environments, from small office networks to large enterprise data centers, campus networks, and even service provider backbones. Its primary function is to dynamically calculate the best path for data packets between routers based on real-

time topology data. OSPF routers continuously share link-state advertisements (LSAs) with each other, enabling them to keep an up-to-date map of the network. This enables quick adaptation to network topology changes, such as the addition or removal of routers and links.

OSPF is used in both single-area and multi-area configurations. In a single-area configuration, all routers in the AS are part of the same OSPF area, and routing information is exchanged freely between all routers. In multi-area OSPF configurations, networks are divided into smaller, more manageable areas to improve scalability and reduce overhead. Routers that belong to different areas exchange summarized routing information, keeping the routing table sizes smaller and more manageable.

Lab Commands

Internal

OSPF

IPv4

```
router ospf [process-id]
  router-id [router-id]
  redistribute connected
  redistribute bgp [AS number]
  network [network] [wildcard mask] area [area ID]
```

IPv6

```
ipv6 router ospf [process-id]
  router-id [router-id]
  redistribute connected
  redistribute bgp [AS number]
interface [interface]
  ip address [ipv4 address] [subnet]
  negotiation auto
  ipv6 address [ipv6 address]
  ipv6 ospf [process-id] area [area number]
  ipv6 ospf hello-interval [hello timer]
  ipv6 ospf dead-interval [dead timer]
```

EIGRP

IPv4

```
router eigrp [EIGRP AS number]
  metric weights 0 0 1 0 0 0
  default-metric [bandwidth] [delay] [reliability] [load] [MTU]
  network [network]
  redistribute bgp [BGP AS number]
  redistribute connected
```

```
passive-interface Loopback0
```

IPv6

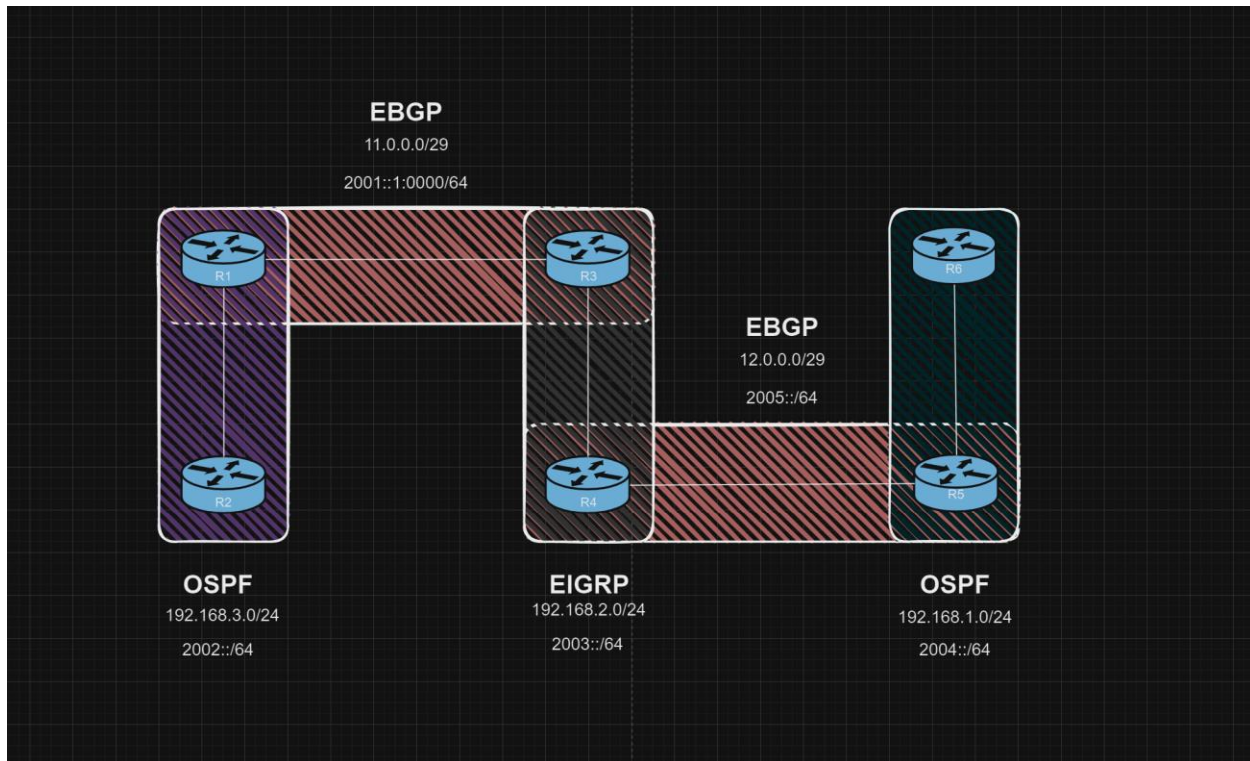
```
ipv6 router eigrp [EIGRP AS number]
  passive-interface Loopback0
  redistribute bgp [BGP AS number]
  default-metric [bandwidth] [delay] [reliability] [load] [MTU]
interface [interface]
  ip address [ipv4 address] [subnet mask]
  negotiation auto
  ipv6 address [ipv6 address]
  ipv6 enable
  ipv6 eigrp [EIGRP AS number]
```

External

EBGP

```
router bgp [BGP AS number]
  bgp router-id [BGP router-id]
  bgp log-neighbor-changes
  neighbor [ipv4 address] remote-as [BGP AS number]
  neighbor [ipv6 address] remote-as [BGP AS number]
  address-family ipv4
    redistribute connected
    redistribute eigrp [EIGRP AS number]
    neighbor [ipv4 address] activate
    no neighbor [ipv6 address] activate
  exit-address-family
  address-family ipv6
    redistribute connected
    redistribute eigrp [EIGRP AS number]
    neighbor [ipv6 address] activate
  exit-address-family
```

Network Diagram



Configurations

R1

```

Current configuration : 4541 bytes
  Last configuration change at 22:23:30 UTC Mon Nov 4 2024
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R1
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
  address-family ipv4
  exit-address-family
  address-family ipv6
  exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-2189345785
  enrollment selfsigned
  subject-name cn=IOS-Self-Signed-Certificate-2189345785
  revocation-check none
  rsa-keypair TP-self-signed-2189345785
crypto pki certificate chain TP-self-signed-2189345785
  certificate self-signed 01

```



```

30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 32313839 33343537 3835301E 170D3234 31313034 32313434
30335A17 0D333030 31303130 30303030 305A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32 31383933
34353738 35308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201 010088EC 75C9C67A CBB1FCD0 1F6DD608 6BAEF6F9 0735893C 11FACADE
D55424AD BA772010 E8A8ACB0 91F0DA4C 48832EDF 332CAA02 5CAA27DE B64D2D80
A50D0AF3 4CEDDDA9 4553E214 3E0A7E5E 1EB7EA48 6D1278B8 C06EE6C7 30D3ED73
CC8EDD91 B4B2DBA5 37DF0E33 B36B9553 D7A10E61 8297D936 838AE700 7FCA5858
3142175C B199ED20 898543BE 50DC3EAF 5694C763 374C3C44 FA7BA937 DECB72AE
6129D3C2 AFA5E357 C2E9846E 259E26DA 53178CBE 38663C59 D81F15D0 AF8D10C6
9B3C4EC1 1961289F 1F152B45 E1009367 DF654080 F1DA3A88 12E3AFEA 1AAFE75
F9F5B6B7 DF738EED 6ABCAF32 D5037348 56CBD528 OCC62784 3DCFF9D2 E07F453A
99794919 9DD70203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603 551D2304 18301680 14317D18 B18715D2 0DEF1477 028CD3CD 6B79596D
A8301D06 03551D0E 04160414 317D18B1 8715D20D EF147702 8CD3CD6B 79596DA8
300D0609 2A864886 F70D0101 05050003 82010100 4875DD88 4F4F795D 1D175CF5
EF443CD1 C9BB83CB 8ED61352 65C76755 43148E49 E6C4AB63 A1C5FC7B 69C43D35
EFFD8CC7 2866809A A8B6672B 621C3A3E 9A837F23 BA4D61B9 A8F6EEC1 FBF885C3
16EEBDED C5E26D43 CB5D1A8E 24ADE6AD 02CC09A3 89C3A472 BEB00BF1 57B128FE
DC655CDC 2ADF05D9 8247005F E8409B9F 6236B2EF B6E210F8 C8EA89B3 B3B7457F
F5BA506A 2BBC6D8E C1D03BD6 96D8181E 00D22C99 F1700E9E 13A1D467 6C4A4EE1
A76C1E90 C3F2A933 4AAC3F99 52A80A9A 05077876 ACEFA91A 25415ED5 572681D3
4A8615A0 6B8B1F5E E7505F4C 97010975 89C6597A 41603373 7AAA008C A069B867
90A18064 D7FF7FD6 06D78114 2CA76107 37CDF1AD
quit
license udi pid ISR4321/K9 sn FDO21482DXE
license boot level appxk9
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface Loopback0
ip address 1.1.1.1 255.255.255.255
interface GigabitEthernet0/0/0
ip address 11.0.0.1 255.255.255.248
negotiation auto
ipv6 address 2001::1:1/64
interface GigabitEthernet0/0/1
ip address 192.168.3.1 255.255.255.0
negotiation auto
ipv6 address 2002::1/64
ipv6 ospf 1 area 1
ipv6 ospf hello-interval 20
ipv6 ospf dead-interval 60
interface Serial0/1/0
interface Serial0/1/1
interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto
interface GigabitEthernet0/2/1
no ip address
shutdown
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 1
router-id 1.1.1.1
redistribute connected subnets
redistribute bgp 1 subnets
network 192.168.3.0 0.0.0.255 area 1
router bgp 1
bgp router-id 1.1.1.1
bgp log-neighbor-changes
neighbor 11.0.0.3 remote-as 2

```

```

neighbor 2001::1:3 remote-as 2
address-family ipv4
    redistribute connected
    redistribute ospf 1
    neighbor 11.0.0.3 activate
    no neighbor 2001::1:3 activate
exit-address-family
address-family ipv6
    redistribute connected
    redistribute ospf 1
    neighbor 2001::1:3 activate
exit-address-family
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
    router-id 1.1.1.1
    redistribute bgp 1
    redistribute connected
control-plane
line con 0
    transport input none
    stopbits 1
line aux 0
    stopbits 1
line vty 0 4
    login
end

```

R2

```

Current configuration : 4021 bytes
  Last configuration change at 21:54:17 UTC Mon Nov 4 2024
  version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R2
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
  address-family ipv4
  exit-address-family
  address-family ipv6
  exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-2557841031
  enrollment selfsigned
  subject-name cn=IOS-Self-Signed-Certificate-2557841031
  revocation-check none
  rsa-keypair TP-self-signed-2557841031
crypto pki certificate chain TP-self-signed-2557841031
  certificate self-signed 01
    30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
    31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
    69666963 6174652D 32353537 38343130 3331301E 170D3234 31313034 32313132
    35365A17 0D333030 31303130 30303030 305A3031 312F302D 06035504 03132649
    4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32 35353738
    34313033 31308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
    0A028201 0100E1CD A7F0E955 1EA1BE12 FF99EB53 3A7BCCFB CB90A7E3 FC135AA8
    83479BEE 4E33388B B32985E9 281289DF 37518CE0 5DDBE686 2D15DDDE 79235E8E
    84F36254 C853982A 3F539A78 C5F3C42E 371942D9 45D8EA8F A089B019 D6CB4B46

```

```

531F426E 358DD6DA EB0767E9 07896C5F EC1C6BE5 A463D39B 01A1E6FD 4981FE8C
528A359A 7BF76F28 48ABAAFE CC17D6DE 4E41153C E3B24C12 83114DEF FDA4AF0B
29150D1C FDFE029F 0F118670 8E17729D A78B6771 2D54F5FD A268CFF0 ADFA0F1E
1A4564F3 A739A6AD A6F976A2 78E34958 1F5B1DF6 8FD69474 5E91F462 6F29C1DF
098D505F 9E101519 48DFDFAE 3EFFFF06 ED554FAD 8F526B42 5A7C9A97 E83447CC
F9DA9193 51950203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603 551D2304 18301680 140FC716 31EAC1A9 83F4C5AE E360507C 47EC69C7
88301D06 03551D0E 04160414 0FC71631 EAC1A983 F4C5AEE3 60507C47 EC69C788
300D0609 2A864886 F70D0101 05050003 82010100 20EDE762 DC30C4BF 446903F4
C6504978 9469C3A0 D256BFC4 78F12705 4D71FE58 8D4E8067 B376AD4B D0E308CE
798501F5 F38D39E3 F8887545 2C17CDF8 B0DF6ADA 0A53113F C0BA020C 3B762509
B0031918 CB45D173 BE4E2909 E680317A 9DC2B0A1 307873AD FE295E4A D8738311
98ED5731 239CFFE4 7F5098E0 B95A1895 228E8319 86FB693E C4A691C3 5D089313
2BE7D1A4 248C9E38 9230CEF3 385FE121 EC311113 1A6A0C0C A3B44873 72345530
C1DE5F7F 800919BD EE6AE63F DCFF2E1D 218896D5 4ADBA12E 82211CE0 733286A5
0B4FC75D 5CA4A4C1 6E2F4CF5 F91D754B D15C0F69 E1BF707C F5408E04 3159471A
600F20AC F7964AAC 158334E5 2D3E4235 60363260
quit
license udi pid ISR4321/K9 sn FDO21500G1N
license boot level appxk9
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
no ip address
negotiation auto
interface GigabitEthernet0/0/1
ip address 192.168.3.2 255.255.255.0
negotiation auto
ipv6 address 2002::2/64
ipv6 ospf 1 area 1
ipv6 ospf hello-interval 20
ipv6 ospf dead-interval 60
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto
interface GigabitEthernet0/2/1
no ip address
shutdown
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 1
router-id 2.2.2.2
network 192.168.3.0 0.0.0.255 area 1
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
router-id 2.2.2.2
control-plane

```

```
line con 0
  transport input none
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login
end
```

R3

```
Current configuration : 4588 bytes
  Last configuration change at 22:02:47 UTC Mon Nov 4 2024
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R3
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
  address-family ipv4
    exit-address-family
  address-family ipv6
    exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-318861592
  enrollment selfsigned
  subject-name cn=IOS-Self-Signed-Certificate-318861592
  revocation-check none
  rsa-keypair TP-self-signed-318861592
crypto pki certificate chain TP-self-signed-318861592
  certificate self-signed 01
    3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
    30312E30 2C060355 04031325 494F532D 53656C66 2D536967 6E65642D 43657274
    69666963 6174652D 33313838 36313539 32301E17 0D323431 31303432 31333332
    365A170D 33303031 30313030 30303030 5A303031 2E302C06 03550403 1325494F
    532D5365 6C662D53 69676E65 642D4365 72746966 69636174 652D3331 38383631
    35393230 82012230 0D06092A 864886F7 0D010101 05000382 010F0030 82010A02
    82010100 9D8597CF C4C868F9 102D46FA B45EF300 23FDD5D2 620AEF06 32185B63
    D3F4F12E 5866AB0E E97041C9 E72781DD 19B90A6F D9064216 AFBF6197 71DE791B
    1ADD331E 53137095 44D19C58 B33C1137 094F2196 86B37BB9 44DD622C 5F58D4C6
    86B8F2BC 01E03B0F 1064F187 371D6812 72C7115D 33AEA33D 2C9A2B8A 481D509E
    70AB6F25 917640E6 92A0E54F 24B86CAB 60B59A57 A10D03C0 BC86D6F5 F385D641
    ED8A2EE4 BD9DF864 BF4FB382 95121F50 0B250523 2A8F2BA6 038B3DDE A8B0F1B2
    2353AC29 7BD76232 877C2B53 417AA95B 6C0B53EA B49200FD 8363869E 26447DED
    5B7A49EC D2F396EF 50C75DA3 68FED164 1CD439B1 37DD952C 053D8596 BF8C429F
    B3434F7F 02030100 01A35330 51300F06 03551D13 0101FF04 05300301 01FF301F
    0603551D 23041830 1680144B 4DFCF9E3 0E108DAA EAD51037 3C6D072B 31D1AD30
    1D060355 1D0E0416 04144B4D FCF9E30E 108DAAEA D510373C 6D072B31 D1AD300D
    06092A86 4886F70D 01010505 00038201 01002203 1F816CF5 056DBB44 65CD79A5
    0DD6BA75 490E9296 2E7FEBC7 23305947 420D3656 6D4A1445 AD52A5D8 4D4FC1D1
    AA63944A 81654C3F 112B2E14 C473C17D 030FEEFE EBD2BDD8 72A8DF62 83093D83
    3AAD9B8B DAC9DC2B 95F30C28 3234B3B6 D292F250 DD8F4657 4DF33859 82C14ED1
    A0D61DBB B529DFE1 C109EE3D 6D8A927D AE214DCC 9C2C5241 4E772877 650C7D35
    06BEDA8F 14264D8E 9645D91C 8E04ABF0 7186B58D D4730A88 05F4531D 3781F9E8
    0CC705D1 EA67ACC7 58BFE880 6D909A62 7DB003CF 136542F9 0888C515 9663DC82
    C407E14C BC1C8162 0181779F 7FB2E5A5 B23B14FB B255774E B0E03604 87AAF87F
    F5A96769 6FDBCBCD 41E9C549 9D8E802F 4875
  quit
license udi pid ISR4321/K9 sn FDO214420HM
no license smart enable
diagnostic bootstrap level minimal
```

```
spanning-tree extend system-id
redundancy
mode none
interface Loopback0
 ip address 3.3.3.3 255.255.255.255
interface GigabitEthernet0/0/0
 ip address 11.0.0.3 255.255.255.248
 negotiation auto
 ipv6 address 2001::1:3/64
interface GigabitEthernet0/0/1
 ip address 192.168.2.1 255.255.255.0
 negotiation auto
 ipv6 address 2003::1/64
 ipv6 enable
 ipv6 eigrp 1
interface Serial0/1/0
 no ip address
 shutdown
interface Serial0/1/1
 no ip address
 shutdown
interface GigabitEthernet0/2/0
 no ip address
 shutdown
 negotiation auto
interface GigabitEthernet0/2/1
 no ip address
 shutdown
 negotiation auto
interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 shutdown
 negotiation auto
router eigrp 1
 metric weights 0 0 1 0 0 0
 default-metric 10000 100 255 1 1500
 network 192.168.2.0
 redistribute bgp 2
 redistribute connected
 passive-interface Loopback0
router bgp 2
 bgp router-id 3.3.3.3
 bgp log-neighbor-changes
 neighbor 11.0.0.1 remote-as 1
 neighbor 2001::1:1 remote-as 1
 address-family ipv4
  redistribute connected
  redistribute eigrp 1
  neighbor 11.0.0.1 activate
  no neighbor 2001::1:1 activate
 exit-address-family
 address-family ipv6
  redistribute connected
  redistribute eigrp 1
  neighbor 2001::1:1 activate
 exit-address-family
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router eigrp 1
 passive-interface Loopback0
 redistribute bgp 2
 default-metric 10000 100 255 1 1500
control-plane
line con 0
 transport input none
 stopbits 1
line aux 0
```



```
stopbits 1
line vty 0 4
login
end
```

R4

```
Current configuration : 4434 bytes
  Last configuration change at 22:56:50 UTC Mon Nov 4 2024
  version 16.9
  service timestamps debug datetime msec
  service timestamps log datetime msec
  platform qfp utilization monitor load 80
  platform punt-keepalive disable-kernel-core
  hostname R4
  boot-start-marker
  boot-end-marker
  vrf definition Mgmt-intf
    address-family ipv4
    exit-address-family
    address-family ipv6
    exit-address-family
  no aaa new-model
  login on-success log
  subscriber templating
  vtp domain cisco
  vtp mode transparent
  ipv6 unicast-routing
  multilink bundle-name authenticated
  crypto pki trustpoint TP-self-signed-2240717686
    enrollment selfsigned
    subject-name cn=IOS-Self-Signed-Certificate-2240717686
    revocation-check none
    rsakeypair TP-self-signed-2240717686
  crypto pki certificate chain TP-self-signed-2240717686
    certificate self-signed 01
      30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
      31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
      69666963 6174652D 32323430 37313736 3836301E 170D3234 31313034 32313235
      30325A17 0D333030 31303130 30303030 305A3031 312F302D 06035504 03132649
      4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32 32343037
      31373638 36308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
      0A028201 0100E608 04ED6277 53AFB217 7E7A1875 2BF73578 32EB33E6 6469055F
      1D6644BA 7CFBA89C BDFB96B2 F595B6CC CD2DF7E0 5A4364A7 C74F08F0 EEDE99AB
      280C2191 FEC81788 BADD4EF3 6905C5C3 7FF09557 82440BB6 8E37EACD 61C5B3AE
      58D89B1B 959C0716 F922AE7A A06BA18A 3BA76957 E03E34D9 34AA10C2 2F2B5FA2
      F4621917 8C90F5D9 52520897 CB330C73 28F7FA5B 8D880563 BF5C5C6E 7E99E0E0
      F5E7D543 F9E7FBE5 66B03D84 894C030E 71A660E9 A0C47C99 F82A6820 55F8DBFD
      0F1579BC 47ECB381 453B3D67 1D11ADF3 58A9723E EF8469B9 87F10ADE 7BEBD744
      B933FF96 9A487E34 14905E87 C7237F57 5EEF8215 D0487AA8 0D43B27A 2557F104
      8905A260 E5A10203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
      301F0603 551D2304 18301680 14EAD6DA CB179878 175C8B6A 1F2E95E5 97EB470E
      C5301D06 03551D0E 04160414 EAD6DACB 17987817 5C8B6A1F 2E95E597 EB470EC5
      300D0609 2A864886 F70D0101 05050003 82010100 C44B2A82 71CC8CFE 181BBAB9
      FAA66ADC 3C10896C 71E330CB 176B517F EF33C108 7C6ED8D7 B5C1D219 3C371D89
      C95AE00E 0B47607A 5C0A4713 700C1258 923A730A 08E8400B 3F051BB2 18694B05
      65C09607 8B41E4FD 3041689F 23411AA0 5E72803A 7BD3AC9F F1DC3594 F5EAB919
      2EA80811 42045E95 79F62CD4 FF8A08C4 75688FC1 E08EA7EE 4592C7F4 309141A1
      78FED3F2 21AED6B4 194E48AD FC13E69C 9C1FD9D4 BD19D2E5 80CFF51B 29779CDE
      06C6E317 EE0B2274 A1F7CAE2 DE62A9F5 9B708DB6 743EF3E5 533ECC5E E6A951C7
      6DA5F515 D15FC3CE CE91D958 C53FBDB0 F439DFE5 E2886730 8D56C9A6 D8E49A6F
      611256C0 6DD82CED 6ED741CD 8232A944 7F20BD12
    quit
  license udi pid ISR4321/K9 sn FDO214414DZ
  no license smart enable
  diagnostic bootup level minimal
  spanning-tree extend system-id
  redundancy
    mode none
  interface Loopback0
    ip address 4.4.4.4 255.255.255.0
```

```
interface GigabitEthernet0/0/0
 ip address 12.0.0.1 255.255.255.248
 negotiation auto
 ipv6 address 2005::1/64
interface GigabitEthernet0/0/1
 ip address 192.168.2.2 255.255.255.0
 negotiation auto
 ipv6 address 2003::2/64
 ipv6 enable
 ipv6 eigrp 1
interface Serial0/1/0
 no ip address
 shutdown
interface Serial0/1/1
 no ip address
 shutdown
interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 shutdown
 negotiation auto
router eigrp 1
 metric weights 0 0 1 0 0 0
 default-metric 10000 100 255 1 1500
 network 192.168.2.0
 redistribute bgp 4
 redistribute connected
 passive-interface Loopback0
router bgp 4
 bgp router-id 4.4.4.4
 bgp log-neighbor-changes
 neighbor 12.0.0.2 remote-as 3
 neighbor 2005::2 remote-as 3
 address-family ipv4
  redistribute connected
  redistribute eigrp 1
  neighbor 12.0.0.2 activate
  no neighbor 2005::2 activate
 exit-address-family
 address-family ipv6
  redistribute connected
  redistribute eigrp 1
  neighbor 2005::2 activate
 exit-address-family
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router eigrp 1
 passive-interface Loopback0
 redistribute bgp 4
 default-metric 10000 100 255 1 1500
control-plane
line con 0
 transport input none
 stopbits 1
line aux 0
 stopbits 1
line vty 0 4
 login
end
```

R5

```
Current configuration : 4406 bytes
Last configuration change at 22:47:13 UTC Tue Nov 5 2024
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
```

```

platform punt-keepalive disable-kernel-core
hostname R5
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
  address-family ipv4
  exit-address-family
  address-family ipv6
  exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-4153289585
  enrollment selfsigned
  subject-name cn=IOS-Self-Signed-Certificate-4153289585
  revocation-check none
  rsakeypair TP-self-signed-4153289585
crypto pki certificate chain TP-self-signed-4153289585
  certificate self-signed 01
    30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
    31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
    69666963 6174652D 34313533 32383935 3835301E 170D3234 31313035 32323035
    31375A17 0D333030 31303130 30303030 305A3031 312F302D 06035504 03132649
    4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D34 31353332
    38393538 35308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
    0A028201 0100B570 BEA378B6 9E112930 66A3F55C 650E587D DA9EBC4F 12C82B7F
    C9BB9010 47C65745 2EBBE04B B82695C6 FFC81D92 233AA846 6A798EE0 DC8EE80F
    F912EB14 08EB3174 4A62B813 6B4608C8 4E81B368 FFB9DC32 ECEDA44F DB42808C
    0A6EA1D9 D2F3750B 5FCD8094 AE29DDBD 20F6FEA0 BF522897 4EB477BB D7C66E96
    E725BF05 7E97AF47 F8BFC41E 0405AE28 E0C36CBA C51C1C34 B6E59BAA A07941CA
    66C30BC4 D856EDC7 74B95BC5 60D9218F 1C429912 8281AEBF AA211B4C A4605FEC
    8C2F3081 F6798A19 CD4E7568 021E5AD4 C10BBAD4 22FFB3AB 447568B4 123E5EDC
    1CE33D89 3DC7A37B 9E1FA3C2 CA5B2D79 333070E1 3417BE69 D2DD9711 A7131F4B
    46C49816 0EB30203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
    301F0603 551D2304 18301680 14CAC4EA 570F3EC9 B1F7D9C5 19B437EC 6DE8D4B5
    36301D06 03551D0E 04160414 CAC4EA57 0F3EC9B1 F7D9C519 B437EC6D E8D4B536
    300D0609 2A864886 F70D0101 05050003 82010100 69C26B03 F35126BB 677EBF0B
    E4819843 76F4436D DF6FE0E8 167562FC C83C4D85 4519DE0B FE07B5CC 4AC981C9
    873B16F5 62EA3248 C5F7711D C82FF648 C7019D9D 411FC9AB 416873F7 5150AFCC
    030F23FA AF1501BB 3EDC1C07 5E4404AB 6FCF804C BCAA221B 327072D5 7E33F299
    EFB334FE 93571C76 D4DC811C 4840E0C5 12565922 043B80FD BB399EFB 0D12CB2C
    358489C2 F0CADCC2 E6D6DBD6 9D1C6A9C C49A4830 7C22FA34 98BFE5E0 2076D86A
    166C81C2 83DB43FD 1EE8DBDF 2B9B5C33 0BFCA27 CBD4214A A741DB84 4A1EBCCE
    D57831AD 00F83953 3724239B F5CF237D 16DC19BC 064BC88F 4CD3A5DD 2D918794
    423E26E2 6231AEC8 0EE72C63 A1FAF0FC BFFFE83B
  quit
license udi pid ISR4321/K9 sn FDO214417Q4
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
  mode none
interface Loopback0
  ip address 5.5.5.5 255.255.255.0
  ip ospf 1 area 1
interface GigabitEthernet0/0/0
  ip address 12.0.0.2 255.255.255.248
  negotiation auto
  ipv6 address 2005::2/64
interface GigabitEthernet0/0/1
  ip address 192.168.1.1 255.255.255.0
  negotiation auto
  ipv6 address 2004::1/64
  ipv6 ospf 1 area 1
  ipv6 ospf hello-interval 20
  ipv6 ospf dead-interval 60
interface Serial0/1/0

```

```
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 1
router-id 1.1.1.1
redistribute connected subnets
redistribute bgp 3 subnets
network 192.168.1.0 0.0.0.255 area 1
router bgp 3
bgp router-id 5.5.5.5
bgp log-neighbor-changes
neighbor 12.0.0.1 remote-as 4
neighbor 2005::1 remote-as 4
address-family ipv4
redistribute connected
redistribute ospf 1
neighbor 12.0.0.1 activate
no neighbor 2005::1 activate
exit-address-family
address-family ipv6
redistribute ospf 1
redistribute connected
neighbor 2005::1 activate
exit-address-family
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
router-id 5.5.5.5
redistribute connected
redistribute bgp 3
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

R6

```
Current configuration : 1761 bytes
Last configuration change at 21:26:10 UTC Fri Nov 1 2024
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R6
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
```

```

multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-2054344109
  enrollment selfsigned
  subject-name cn=IOS-Self-Signed-Certificate-2054344109
  revocation-check none
  rsa-keypair TP-self-signed-2054344109
crypto pki certificate chain TP-self-signed-2054344109
license udi pid ISR4321/K9 sn FDO215009QY
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
  mode none
interface Loopback0
  ip address 6.6.6.6 255.255.255.0
  ip ospf 1 area 0
interface GigabitEthernet0/0/0
  no ip address
  negotiation auto
interface GigabitEthernet0/0/1
  ip address 192.168.1.2 255.255.255.0
  negotiation auto
  ipv6 address 2001::2:2/64
  ipv6 ospf network point-to-point
interface Serial0/1/0
  no ip address
interface Serial0/1/1
  no ip address
interface GigabitEthernet0
  vrf forwarding Mgmt-intf
  no ip address
  negotiation auto
router ospf 1
  router-id 2.2.2.2
  redistribute connected subnets
  network 192.168.1.0 0.0.0.255 area 1
ip forward-protocol nd
no ip http server
ip http secure-server
ip tftp source-interface GigabitEthernet0
control-plane
line con 0
  transport input none
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login
end

```

Lab

Enable IPv6 routing across all routers using the command `[ipv6 unicast-routing]`.

```
ipv6 unicast-routing
```

Proceed to apply IP addresses according to the topology. As ipv6 routing has been enabled the configuration also includes the IPv6 addresses.

```
interface GigabitEthernet0/0/0
 ip address 12.0.0.1 255.255.255.248
 negotiation auto
 ipv6 address 2005::1/64
```

BGP

For this configuration, following the topology above routers 1-3 and 4-5 will use eBGP.

To initialize BGP you must assign it an autonomous system number. Then manually configure a router its id.

For BGP to function there must also be a router-id and neighbor definitions.

To define neighbors, use the command [*neighbor ip remote-as autonomous system number*] to direct neighbors.

```
router bgp 4
 bgp router-id 4.4.4.4
 bgp log-neighbor-changes
 neighbor 12.0.0.2 remote-as 3
 neighbor 2005::2 remote-as 3
```

For interconnecting Ipv4 BGP must receive the commands [*redistribute protocol AS/process-id*]. Along with activating neighbors.

```
address-family ipv4
 redistribute connected
 redistribute eigrp 1
 neighbor 12.0.0.2 activate
 no neighbor 2005::2 activate
 exit-address-family
```

Ipv6 is not much different with activating and redistributing protocols. However, you must use [*address-family ipv6*] in the router BGP configuration.

```
address-family ipv6
  redistribute connected
  redistribute eigrp 1
  neighbor 2005::2 activate
exit-address-family
```

OSPF

To activate ospf use the command `[router ospf process-id]` then configure the router-id, whether automatically or manually. Finally for OSPF Ipv4 it requires the command `[network ipv4 wildcard mask area area]` to determine the network to attempt to get neighbors in. OSPF also requires redistribute commands to redistribute routes learned from other protocols.

```
router ospf 1
  router-id 1.1.1.1
  redistribute connected subnets
  redistribute bgp 3 subnets
  network 192.168.1.0 0.0.0.255 area 1
```

For IPv6 configuration OSPF may be the most different of the protocols used in this configuration. To activate OSPFv3 it requires the ipv6 notation before router. The main difference being to not use the network statement here but rather individually in interfaces. Be sure to redistribute again along with the configuration of hello and dead timers.

```
ipv6 router ospf 1
  router-id 5.5.5.5
  redistribute connected
  redistribute bgp 3
```

```
interface GigabitEthernet0/0/1
  ip address 192.168.1.1 255.255.255.0
  negotiation auto
  ipv6 address 2004::1/64
  ipv6 ospf 1 area 1
  ipv6 ospf hello-interval 20
  ipv6 ospf dead-interval 60
```


EIGRP

EIGRP also uses autonomous system numbers, to enable EIGRP on a router use the command [router eigrp AS]. As EIGRP is a metric based routing protocol the [default-metric bandwidth delay load mtu reliability] command is required for redistributing routes and for EIGRP to function correctly along with the [metric weights k1 k2 k3 k4 k5] to determine k routes.

```
router eigrp 1
metric weights 0 0 1 0 0 0
default-metric 10000 100 255 1 1500
network 192.168.2.0
redistribute bgp 4
redistribute connected
passive-interface Loopback0
```

EIGRP for IPv6 will be roughly the same, with the same default metric command and metric weights however, just with different IPv6 addresses. Do not forget to redistribute the other protocol on the router.

```
ipv6 router eigrp 1
passive-interface Loopback0
redistribute bgp 4
default-metric 10000 100 255 1 1500
```

Problems

After initial configurations sometimes routes will not be established between different areas; the fix for the configuration above was the missing 'redistribute' command on IPv6 processes. Every process whether IPv4 or IPv6 requires the 'redistribute' command to a neighboring router to successfully share routes between the routing tables.

Conclusion

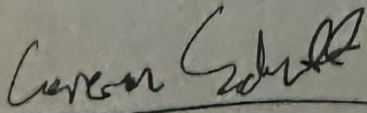
eBGP allows for bordering ISP's and networks to communicate with one another on their internal or external networks. External BGP is a bridge between protocols long distance and is the only viable option for long distance routing. Using AS numbers for identification and neighbor statements to stay connected even through different protocols and networks. With all the little precision internal routing (commonly) done by OSPF or EIGRP.

Sign Off

Cisco CCNP Sign off Sheet

Notes/Issues and Stamp area below. Lab: eBGP and cross network routing





Name: Cameron Schaff Date: 24-11-18

